



NATIONAL INSTITUTE OF TECHNOLOGY
ROURKELA – 769 008

Department	Computer Science and Engineering
Examination	Mid Semester – 2025-26
Program	B. Tech / M. Tech / Ph. D
Session	Spring
Course code & Course Name	CS6404 - Software Testing
Marks	30
Duration	2 Hours

This question paper contains 2 pages

Instructions:

Answer all questions.

Q. No.	Questions	Marks	CO*	BL**
1.	Design a black-box test suite for a program that accepts a pair of points defining a straight line and another point and a floating point number defining the center of a circle and its radius. The program is intended to compute their points of intersection and prints them.	5	CO-1	BL-4
2.	Consider a scenario in which we need to derive the test cases for a Library Information System, that has four different functionalities as mentioned below: (i) Query book (ii) Issue book (iii) Return book (iv) Reserve book These four functionalities can individually be <i>available</i> or <i>unavailable</i> or <i>partially available</i> to the users. Design test cases for the above Library Information System using orthogonal array testing. Show all the steps.	5	CO-1	BL-3
3.	Consider the following program. <pre>#include <stdio.h> main{ int a,b,c,d; clrscr(); printf("Enter the two variables a, b"); scanf("%d %d", &a , &b); printf("Enter the option 1:Addition, 2:Subtraction, 3:Multiplication, 4:Division"); scanf("%d",&c); switch(c){ case1:d= a+b; printf("Addition of the two no.=%d",d); break; case2:d= a-b; printf("Subtraction of the two no.=%d",d); break; case3:d= a*b; printf("Multiplication of the two no.=%d",d); break; case4:d= a/b; printf("Division of the two no.=%d",d); break; } }</pre>	5	CO-2	BL-3

	i. Draw the CFG for the above program. ii. Calculate the cyclomatic complexity of the program. iii. List all the linearly independent paths. iv. Design all the test cases from the independent paths.			
4.	What do you mean by concolic testing? Design test cases for the following code using concolic testing. Show all the intermediate steps. <pre> x = input (); y = input (); If (x < y) { Assert (x != 200); } </pre>	5	CO-2	BL-3
5.	Perform top-down integration procedure for the following system hierarchy. Show all the intermediate steps. Compute the effort required for this. <pre> graph TD 1[1] --> 2a[2a] 1 --> 2b[2b] 2a --> 3[3] 2a --> 4[4] 2a --> 5[5] 2a --> 6[6] 2a --> 7[7] 2a --> 8[8] 2b --> 7 2b --> 8 3 --> 9[9] 3 --> 10[10] 4 --> 11[11] 5 --> 12[12] 6 --> 13[13] 7 --> 14[14] 7 --> 15[15] 8 --> 16[16] </pre>	5	CO-3	BL-4
6.	Suppose you have to develop a software to automate the student admission process of NIT Rourkela. Explain the different types of performance (non-functional) testing that you will perform for the above software. You may assume your own functional and non-functional requirements.	5	CO-3	BL-3

CO**: Course Outcome; *BL**: Bloom's Taxonomy Level.